



Overview of the CLEF 2024 SimpleText Track

Improving Access to Scientific Texts for Everyone

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Abstract. Everyone acknowledges the importance of objective scientific information. However, finding and understanding relevant scientific documents is often challenging due to complex terminology and readers' lack of prior knowledge. The question is can we improve accessibility for everyone? This paper presents an overview of the SimpleText Track at CLEF 2024 addressing the technical and evaluation challenges associated with making scientific information accessible to a wide audience, including students and non-experts. It describes the data and benchmarks provided for scientific text summarization and simplification, along with the participants' results. The CLEF 2024 SimpleText track is based on four interrelated tasks: Task 1 on *Content Selection*: Retrieving Passages to Include in a Simplified Summary. Task 2 on *Complexity Spotting*: Identifying and Explaining Difficult Concepts. Task 3 on *Text Simplification*: Simplify Scientific Text. Task 4 on *SOTA?*: Tracking the State-of-the-Art in Scholarly Publications.

Keywords: Scientific text simplification · Information extraction · Information retrieval · Natural language processing

1 Introduction

The importance of objective scientific information is universally acknowledged. In practice, accessing, processing, and comprehending relevant scientific documents is challenging due to complex terminology and the potential lack of prior knowledge among readers. The CLEF 2024 SimpleText track aims at improving accessibility to scientific information for everyone, both in terms of information retrieval and natural language processing. The workshop at CLEF 2021 [12] and tracks at CLEF 2022 [16] and 2023 [15] resulted in research community and test collections for improving access to

scientific information for everyone. Specifically, test collections for retrieving relevant (and accessible) scientific text [34], for simplifying the language used in scientific documents without compromising the accuracy of the information [13], and for making complex concepts more understandable to a broader audience [11].

Scientific Text Simplification is different from traditional text simplification approaches focusing on lower literacy levels, for example making general text accessible to young readers. Recent advances in IR and NLP hold the promise of removing some of the barriers to scientific information access.¹ The overall impact of CLEF SimpleText is to increase science literacy and broaden the audience of objective, scientific information.

The track's setup is based on the following pipeline: i) select the information to be included in a simplified summary; ii) improve the readability of the scientific text; iii) provide additional background knowledge for remaining difficult concepts; and iv) aggregate information from multiple articles. This results in the following four tasks [16]:

- **Task 1: Content Selection** Retrieving Passages to Include in a Simplified Summary.
- **Task 2: Complexity Spotting** Identifying and Explaining Difficult Concepts.
- **Task 3: Text Simplification** Simplify Scientific Text.
- **Task 4: SOTA?** Tracking the State-of-the-Art in Scholarly Publications.

A total of 45 teams registered for our SimpleText track at CLEF 2024. A total of 20 teams submitted 207 runs in total. The statistics for these runs submitted are presented in Table 1. However, some runs had problems that we could not resolve. We do not detail them in the paper as well as the 0-scored runs.

This paper gives an overview of the CLEF 2024 SimpleText Track. Further detail per track is provided in the Track overview papers published in the CEUR CLEF Working Notes, specifically for Task 1 on *Content Selection* [35]; Task 2 on *Complexity Spotting* [31]; Task 3 on *Text Simplification* [17]; and Task 4 on *SOTA?* [9].

In the rest of this paper, we will provide a detailed description of each task of the CLEF 2024 edition in four self-contained sections: Task 1: Content Selection in Sect. 2, Task 2: Complexity Spotting in Sect. 3, Task 3: Text Simplification in Sect. 4, and Task 4: SOTA? in Sect. 5. We end with a discussion and conclusions in Sect. 6.

2 Task 1: Retrieving Passages to Include in a Simplified Summary

This section details *Task 1: Content Selection* on Retrieving Passages to Include in a Simplified Summary.

2.1 Description

Given a popular science article targeted to a general audience, this task aims at retrieving passages, which can help to understand this article, from a large corpus of academic abstracts and bibliographic metadata. Relevant passages should relate to any of the topics in the source article.

¹ A joined effort with others like Scholarly Document Processing <https://sdproc.org/2024/>.

Table 1. CLEF 2024 Simpletext official run submission statistics

Team	Task 1	Task 2			Task 3		Task 4		Total runs
		2.1	2.2	2.3	3.1	3.2	4.1	4.2	
AIIRLab	5	3	3		4	4			19
AMATU							3	9	12
Arampatzis	9	5	5	2	4	4			29
Elsevier	10				8	2			20
L3S							12	12	24
LIA	5								5
PiTheory					11	10			21
Sharigans	1	1	1		1	1			5
SINAI		3	3						6
SONAR					1				1
AB/DPV	1	1	1		1				4
Dajana/Katya		1			1				2
Frane/Andrea		1	1		1				3
Petra/Regina	1	1			1				3
Ruby	1	1			1	1			4
Tomislav/Rowan	2	2			1	1			6
UAmsterdam	6	1		2	4	6			19
UBO	1	1	1		2	2			7
UniPD		3	3						6
UZHPandas					11				11
Total runs	42	24	18	4	52	31	15	21	207

Data. We use popular science articles as a source for the types of topics the general public is interested in and as a validation of the reading level that is suitable for them. The main corpus is a large set of scientific abstracts plus associated metadata covering the fields of computer science and engineering. We reuse the collection of academic abstracts from the Citation Network Dataset (12th version released in 2020)² [37]. This collection was extracted from DBLP, ACM, MAG (Microsoft Academic Graph), and other sources. It includes 4,232,520 abstracts in English, published before 2020.

Search requests are based on popular press articles targeted to a general audience, based on *The Guardian* and *Tech Xplore*. Each of these popular science articles represents a general topic that has to be analyzed to retrieve relevant scientific information from the corpus.

We provide the URLs to original articles, the title, and the textual content of each popular science article as a general topic. Each general topic was also enriched with one or more specific keyword queries manually extracted from their content, creating a familiar information retrieval task ranking passages or abstracts in response to a query. Available training data from 2023 includes 29 (train) and 34 (test) queries, with the later set having an extensive recall base due to the large number of submissions in 2023 [34].

² <https://www.aminer.cn/citation>.

In 2024, we introduced two novelties. First, in addition to the ElasticSearch we provided to participants, we made available a new vector database with sentence embedding scores. This database stores for each article their ID, two sentence-embedding vectors computed by all-MiniLM-L6-v2³ from their title and their abstract. Second, we added between 2 and 5 new queries (with IDs of the form G*.C*) for each of the 20 articles from the Guardian. These topics were generated by ChatGPT 4, with a prompt asking to list the main subtopics related to computer science; they were manually inspected to check they are linked to the original article and are not redundant. They are longer, containing around ten words and focusing on a specific point related to the article. An example of a keyword query is “*system on chip*” (T06.1) and an example of a long query is “*How AI systems, especially virtual assistants, can perpetuate gender stereotypes?*” (G01.C1).

Evaluation. Topical relevance was evaluated with a 0–2 score on the relevance degree towards the content of the original article. We dramatically extended the training qrels with 9,990 additional evaluated documents, with a main focus on the new long queries for the Guardian, and the T06–T11 queries for which no list of relevant documents had yet been provided.

In addition to topical relevance, we took into account other key aspects of the track, such as the text complexity and the credibility of the retrieved results. These evaluations were performed using automatic metrics.

2.2 Participant’s Approaches

A total of 11 teams submitted 42 runs in total.

AB/DPV. Varadi and Bartulović [40] submitted 1 run for Task 1. They used our ElasticSearch API and took into account an FKGL readability score for their combined score.

Sharingans. Ali et al. [2] also submitted 1 run. They experimented with the ColBERT neural ranker and used GPT 3.5 to select the most informative and concise passages for inclusion in the summary.

Tomislav/Rowan. Mann and Mikulandric [27] submitted a total of 2 runs. They took the top 100 results retrieved by ElasticSearch. Then, they used cosine similarity on TF-IDF vectors as the relevance score and FKGL score as the combined score.

Petra/Regina. Elagina and Vučić [10] submitted 1 run, for the first 3 queries only, with the same approach as the previous system.

AIIRLab. Largey et al. [25] submitted a total of 5 runs and proposed several models. First, since input queries are short keyword terms, they used query expansion with LLaMA 3 and reranked the top 5,000 results retrieved by TF-IDF with a bi-encoder or a cross-encoder. Second, they applied LLaMa3 as a pairwise re-ranker. Third, they leveraged ElasticSearch with fine-tuned cross-encoders.

UBO. Vendeville et al. [41] submitted a total of 1 run. They used PyTerrier⁴ to retrieve documents from TF-IDF scores. Then, the MonoT5 reranker provided by PyTerrier was employed to reorder all extracted documents.

³ <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>.

⁴ <https://pyterrier.readthedocs.io/>.

UAmsterdam. Bakker et al. [4] submitted a total of 6 runs for Task 1. First, they focused on regular information retrieval effectiveness with 2 vanilla baseline runs on an Anserini index, using either BM25 or BM25+RM3, and 2 other runs generated with neural cross-encoder rerankings of these runs by an MS MARCO-trained ranker. Second, 2 further runs filter out the most complex abstracts per request, using the median FKGL readability measure.

Elsevier. Capari et al. [5] submitted a total of 10 runs. Their approaches mainly centered on creating a ranking model. They started by assessing the performance of several models on a proprietary test collection of scientific papers. Then, the top-performing model was fine-tuned on a large set of unlabeled documents using the Generative Pseudo Labeling approach. They also experimented with generating new search queries.

LIA submitted a total of 5 runs as baselines for Task 1. All five have been included in the pool of results for q-rel evaluation. The first three runs (elastic, meili, and boolean) used bag-of-words models and sparse vector document representation. One was generated with the Elastic 7 Search API (elastic), another was gathered with the Meilisearch system⁵ (meili) based on bucket sort, and the third one was produced with a simple boolean model (boolean) powered by PostgreSQL GIN text indexing. Two additional runs relied on embedding vectors based on the paragraph cross-encoder MS MARCO Mini LM⁶. These embeddings, along with a search API based on them, have been released to participants. Documents are ranked based on the dot product between the query and the abstract (vir_abstract) or the title (vir_title) using the pg_vector⁷ PostgreSQL extension and an ivfflat dense vector index (k-means vector clustering with $\sqrt{|D|}$ centroids).

Ruby. This team (No paper received) submitted a total of 1 run for Task 1. Their approach relies on Elasticsearch and a TF-IDF score.

Arampatzis. This team (No paper received) submitted a total of 9 runs for Task 1. As these reports are very close, the Tables below only report their evaluation made on their first run.

2.3 Test Results

Table 2 still shows relevance evaluation, with a ranking by NDCG@10 on queries absent from the training qrels. Note that for the sake of brevity, we only consider here the relevance score provided by participants. The selection of top results against the combined score would have modified the relevance evaluation.

2.4 Analysis

We complement these evaluations by taking into consideration other aspects essential for Task 1. Table 3 highlights credibility and text complexity. We used simple automatic metrics to provide an overview of the importance and the complexity of the article. First,

⁵ <https://www.meilisearch.com/>.

⁶ <https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-12-v2>.

⁷ <https://github.com/pgvector/pgvector>.

Table 2. Results for CLEF 2024 SimpleText Task 1 on the Test qrels (G01.C1-G10.C1 and T06-T11).

Run	MRR	Precision		NDCG		Bpref	MAP
		10	20	10	20		
AIIRLab_Task1_LLaMABiEncoder	0.9444	0.8167	0.5517	0.6170	0.5166	0.3559	0.2304
AIIRLab_Task1_LLaMAReranker2	0.9300	0.7933	0.5417	0.5943	0.5004	0.3495	0.2177
AIIRLab_Task1_LLaMAReranker	0.8944	0.7967	0.5583	0.5889	0.5011	0.3541	0.2200
LIA_vir_title	0.8454	0.6933	0.4383	0.5013	0.3962	0.3594	0.1534
AIIRLab_Task1_LLaMACrossEncoder	0.7975	0.6933	0.5100	0.4745	0.4240	0.3404	0.1970
LIA_vir_abstract	0.7683	0.6000	0.4067	0.4207	0.3504	0.3857	0.1603
UAMS_Task1_Anserini_rm3	0.7878	0.5700	0.4350	0.3924	0.3495	0.4010	0.1824
UAMS_Task1_Anserini_bm25	0.7187	0.5500	0.4883	0.3750	0.3707	0.3994	0.1972
UAMS_Task1_CE1K_CAR	0.5950	0.5333	0.4583	0.3672	0.3618	0.2701	0.1605
UAMS_Task1_CE1K	0.5950	0.5333	0.4583	0.3672	0.3618	0.4032	0.1939
UAMS_Task1_CE100_CAR	0.6618	0.5300	0.4567	0.3654	0.3549	0.2657	0.1579
UAMS_Task1_CE100	0.6618	0.5300	0.4567	0.3654	0.3549	0.2657	0.1579
AIIRLAB_Task1_CERRF	0.7264	0.5033	0.4000	0.3584	0.3239	0.2204	0.1309
Arampatzis_1.GPT2_search	0.6986	0.5100	0.2550	0.3516	0.2462	0.0742	0.0577
UBO_Task1_TFIDFT5	0.7132	0.4833	0.3817	0.3474	0.3197	0.2354	0.1274
LIA_bool	0.7242	0.5233	0.3633	0.3381	0.2891	0.2661	0.1199
Elsevier@SimpleText_task1_run8	0.7123	0.4533	0.3367	0.3146	0.2752	0.1582	0.0906
Elsevier@SimpleText_task1_run4	0.6162	0.4300	0.3217	0.3063	0.2681	0.1642	0.1005
Elsevier@SimpleText_task1_run10	0.5117	0.4067	0.2767	0.2885	0.2365	0.1236	0.0729
LIA_elastic	0.6173	0.3733	0.2900	0.2818	0.2442	0.3016	0.1325
AB/DPV_SimpleText_task1_FKGL	0.6173	0.3733	0.2900	0.2818	0.2442	0.1966	0.1078
Ruby_Task1	0.5470	0.4233	0.3533	0.2756	0.2671	0.1980	0.1110
LIA_meili	0.6386	0.4700	0.2867	0.2736	0.2242	0.2377	0.0833
Elsevier@SimpleText_task1_run6	0.5333	0.3833	0.3117	0.2633	0.2430	0.1841	0.0973
Tomislav/Rowan/Rowan_SimpleText_T1_1	0.5444	0.3733	0.2750	0.2443	0.2183	0.0963	0.0601
Elsevier@SimpleText_task1_run5	0.4867	0.3533	0.2883	0.2408	0.2232	0.1834	0.0943
Elsevier@SimpleText_task1_run1	0.5589	0.3000	0.3300	0.2247	0.2399	0.1978	0.1018
Elsevier@SimpleText_task1_run7	0.4026	0.3200	0.2250	0.2168	0.1850	0.1085	0.0565
Elsevier@SimpleText_task1_run9	0.3868	0.3300	0.2283	0.2105	0.1829	0.1103	0.0590
Elsevier@SimpleText_task1_run3	0.4733	0.2367	0.2033	0.1853	0.1703	0.1587	0.0714
Elsevier@SimpleText_task1_run2	0.4193	0.2233	0.2433	0.1803	0.1865	0.1768	0.0820
Sharingans_Task1_marco-GPT3	0.6667	0.0667	0.0333	0.1149	0.0797	0.0107	0.0107
Tomislav/Rowan_SimpleText_T1_2	0.0217	0.0233	0.0150	0.0121	0.0106	0.0062	0.0025
Petra/Regina_simpleText_task1	0.0026	0.0000	0.0050	0.0000	0.0035	0.0031	0.0007

the average number of bibliographic references among the top 10 results of each query is provided. Second, we give the average size of sentences and the average number of syllabi per word in the abstract of these results. Note that this time we considered the

Table 3. Evaluation of complexity and credibility for SimpleText Task 1 (over all 176 queries).

Run	Avg #Refs	Avg sentence length	Avg syllabus per word
AB/DPV_SimpleText_task1s_FKGL	9.7	30.4	1.987
Arampatzis_1.GPT2_searchs	10.5	22.1	1.916
LIA_bool	13.0	33.1	1.906
LIA_elastic	9.2	21.1	1.812
LIA_meili	9.6	23.8	1.740
LIA_vir_abstract	7.2	21.0	1.885
LIA_vir_title	9.8	22.4	1.870
AIIRLAB_Task1_CERRF	10.6	22.0	1.895
AIIRLab.Task1.LLaMABiEncoder	9.5	31.0	1.865
AIIRLab.Task1.LLaMACrossEncoder	10.0	30.6	1.890
AIIRLab.Task1.LLaMAReranker	8.8	22.1	1.772
AIIRLab.Task1.LLaMAReranker2	8.6	20.9	1.707
Elsevier@SimpleText_task_1_run1	10.0	22.2	1.888
Elsevier@SimpleText_task_1_run10	10.2	22.3	1.881
Elsevier@SimpleText_task_1_run2	11.2	22.4	1.893
Elsevier@SimpleText_task_1_run3	9.7	22.0	1.894
Elsevier@SimpleText_task_1_run4	10.7	22.2	1.881
Elsevier@SimpleText_task_1_run5	11.1	22.3	1.886
Elsevier@SimpleText_task_1_run6	11.2	22.5	1.885
Elsevier@SimpleText_task_1_run7	9.8	22.2	1.870
Elsevier@SimpleText_task_1_run8	10.3	22.4	1.903
Elsevier@SimpleText_task_1_run9	10.6	22.2	1.857
Petra/Reginas_simpleText_task_1	5.5	22.1	1.955
Ruby_Task_1	9.6	21.2	1.837
Sharingans.Task1_marco-GPT3	9.8	23.0	1.896
Tomislav/Rowan_SimpleText.T1_1	11.3	24.6	1.952
Tomislav/Rowan_SimpleText.T1_2	11.8	23.6	1.943
UAms_Task1_Anserini_bm25	11.8	23.7	1.893
UAms_Task1_Anserini_rm3	11.9	24.8	1.894
UAms_Task1_CE100	11.1	22.3	1.901
UAms_Task1_CE100_CAR	10.6	19.6	1.832
UAms_Task1_CE1K	10.8	22.4	1.872
UAms_Task1_CE1K_CAR	10.2	19.5	1.809
UBO_Task1_TFIDFT5	10.3	22.2	1.899

comb score to elect the top 10 results, which favors the systems w.r.t. the relevance score which may be an orthogonal criterion.

This concludes the results for the CLEF 2024 SimpleText Task 1: Content Selection on Retrieving Passages to Include in a Simplified Summary. Our main findings are the

following: First, the Tables on relevance are dominated by neural rankers, in particular, cross-encoders and LLaMA 3 used as a pairwise re-ranker. Second, a majority of participants relied on ElasticSearch search results. If neural models used in processing steps leveraged these results, other IR systems turned out to be competitive. For instance, LIA_vir_title operating with embedding sentences or UAMS_Task1_Anserini_rm3, using an Anserini index have high relevance evaluations. Third, as expected, ranking over systems differs according to the considered criterion. Runs filtered against readability measures tend to have shorter sentences with a more or less drop in relevance. Remarkably, LLaMA 3 used as a reranker seems to not only help to select more relevant documents but also with more concise sentences. We refer the reader to the CLEF 2024 SimpleText Task 1 Overview paper [35] for further details and discussion.

3 Task 2: Identifying and Explaining Difficult Concepts

This section details *Task 2: Complexity Spotting* on Identifying and Explaining Difficult Concepts.

3.1 Description

The goal of this task is to decide which concepts in scientific abstracts require explanation and contextualization in order to help a reader understand the scientific text. Since 2023, we have asked participants to identify such concepts and to provide useful and understandable explanations for them. Thus, the task has three subtasks: Task 2.1) to identify candidate terms in a given passage from a scientific abstract, and set the level of difficulty of the concept designated by each term (easy, medium, or difficult); Task 2.2) to provide a definition or an explanation or both only for the difficult terms⁸; Task 2.3) to rank a set of provided definitions for each difficult term.

Data. The corpus of Task 2 is based on the sentences in high-ranked abstracts to the requests of Task 1 collected in 2023 [11]. A total of 175 documents and 1,077 sentences were used to generate the training and test data. In particular, we had 115 documents and 576 sentences for building the training set and 60 documents and 501 sentences for building the test set.

For the training set, we asked 21 experts to manually annotate each training document in order to produce the set of terms for each sentence, the corresponding difficulty, and the definitions and explanations for each difficult term. A total of 1,609 terms and 899 definitions and explanations were generated. In some cases, we intentionally assigned the same documents to different experts in order to have the possibility to further study the agreement between the extraction of terms and the generation of definitions. For each term with a definition, we also generated two “good” definitions and two “bad” definitions in order to create the set of definitions that should be ranked in Task 2.3. A total of 2,356 sentences (equally distributed between good and bad) were generated. In addition to this first set of training data, we also created an additional set of files that were produced by an external expert who reviewed the work of the 21

⁸ Henceforth, we will use ‘difficult term’ to indicate a term that designates a difficult concept.

experts. We called these additional files the validation set. In these files, the external experts added what she thought was missing in the first round of annotation (either terms or definitions or both). An additional set of 677 terms and, 960 definitions, and 3,732 generated definitions (equally distributed between good and bad definitions) were added to the training set.

For the test set, we asked the external expert to annotate the remaining 60 documents. A total of 1440 terms were extracted and 424 definitions were written from the 501 sentences of the test set. An additional 3,816 definitions (equally distributed between good and bad definitions) were also added.

Finally, we encouraged participants to train on existing datasets extracted from other resources such as the WCL dataset [29] to train the definition generation model or use gazetteers, wikification resources as well as resources for abbreviation deciphering.

Evaluation. We evaluated difficult concept spotting, Task 2.1, and their definitions, Task 2.2, in terms of recall, precision, and BLEU score [11]. The ranking of definitions, Task 2.3, will be evaluated with precision@1 and precision@5. A qualitative analysis will also be performed in order to study the problems of term identification and the generation of definitions.

In addition, we will manually evaluate the provided explanations in terms of their usefulness with regard to a query as well as their complexity for a general audience. Note that the provided explanations can have different forms, e.g. abbreviation deciphering, examples, use cases, etc.

3.2 Participant's Approaches

A total of 13 teams submitted 46 runs in total, with many being a combined Task 2.1 and Task 2.2 submission.

AB/DPV. Varadi and Bartulović [40] submitted a total of one run for Task 2. Their main approach was to use natural language processing to extract difficult terms from passages, followed by generating definitions for them or retrieving them from sources such as Wikipedia. The participants did not submit runs on the test set.

AIIRLab. Largey et al. [25] submitted a total of three runs for Task 2. The participants used LLaMA 3 and Mistral language models to create the three runs. The methodology involves prompt engineering and reinforcement learning with human feedback to improve the quality of outputs generated by the LLaMA model.

Dajana/Kathy submitted a total of one run for Task 2. The participants used the LLaMA-2 13B model but they did not provide additional information about the run.

Frane/Andrea submitted a total of one run for Task 2. Participants did not provide additional information about the run.

Sharingans. Ali et al. [2] submitted a total of one run for Task 2. The participants fine-tuned GPT 3.5 turbo model for the selection of difficult terms as well as the generation of definitions and explanations for the extracted terms. Prompt-engineering techniques were employed to construct specific prompts that guided the model in producing accurate and contextually relevant definitions.

SINAI. Ortiz-Zambrano et al. [33] submitted a total of three runs for Task 2. Their approach is to apply learning cues without prior examples to the GPT-4-Turbo model, extracting predictions from the generated sequence. They also used the OpenAI API in Python to interact with the model, allowing for an easy integrate GPT-4-Turbo into their workflow.

Petra/Regina. Elagina and Vučić [10] submitted a total of one run for Task 2. The participants employed a combination of named entity recognition (NER) techniques and rule-based approaches to identify and extract entities such as proteins, genes, and chemical compounds. In particular, they used spaCy for NER and developed custom rules for entity extraction.

Tomislav/Rowan. Mann and Mikulandric [27] submitted a total of two runs for Task 2. The team created a prompt for LLaMA 2 13B model that asked the LLM to iterate over each of their source sentences and extract three scientific terms from each sentence and then were sorted and prompted to LLaMA to return a difficulty rating. Wikipedia was used to return definitions for the difficult terms.

UAmsterdam. Bakker et al. [4] submitted a total of three runs for Task 2. The participants used an IDF-based term weighting to locate the rarest terms for Task 2.1. For Task 2.3, they developed an approach to rank definitions or explanations for a given sentence and term pair looking at the textual similarity of the large set of provided sentences.

UBO. Vendeville et al. [41] submitted a total of one run for Task 2. In particular, the participants used a Small Language Model, Phi3 mini, without fine-tuning with a one-shot prompt.

UniPD. Di Nunzio et al. [8] submitted a total of three runs for Task 2. Their participation in Task 2 focused on identifying and explaining difficult content using Large Language Models (LLMs) to enhance text simplification. The methodology involves iterative experimentation with various prompting strategies to optimize the performance of the model in this task.

Ruby. (No paper received) submitted a single run for Task 2.1. This run can only be evaluated on the train data.

Arampatzis. (No paper received) made 5 submissions for Tasks 2.1 and 2.2 each, and two submissions for Task 2.3. Their Task 2.1 and Task 2.2 submissions only contain results for the train data.

3.3 Results

In this section, we present the results on the test set for Task 2.1 and Task 2.2. At present time, the results for Task 2.3 are still ongoing (with only four runs by two participants) and will be made available in the future.

3.4 Test Results

The results on the test set are summarized in Table 4. For each run, we report:

- the recall of all the terms, independently from the level of difficulty;
- the precision of all the terms, independently from the level of difficulty;
- the recall of the difficult terms;
- the precision of the difficult terms;
- the BLEU score computed for bigrams (ngrams with $n = 2$).⁹

Table 4. Results for CLEF 2024 SimpleText Task 2

runid	recall	precision	recall_difficult	The results shown in	bleu_n2_average
AIIRLab_Task2.2.LLaMA	0.28	0.65	0.26	0.67	0.15
AIIRLab_Task2.2.LLaMAFT	0.01	0.99	0.00	1.00	0.12
AIIRLab_Task2.2.Mistral	0.41	0.69	0.19	0.49	0.13
Dajana/Kathy_SimpleText_Task2.2.LLaMA2.13B.CHAT	0.01	0.59	0.00	0.00	0.00
Frane/Andrea_SimpleText_Task2.2.LLaMA2.13B.CHAT	0.01	0.65	0.01	0.36	0.00
team1_Petra_and_Regina_Task2_ST	0.00	0.50	0.00	0.00	0.00
Sharingans_Task2.2.GPT	0.47	0.55	0.54	0.60	0.10
SINAL.task_2.PRM_ZS_TASK2_V1	0.09	0.82	0.10	0.52	0.16
SINAL.task_2.PRM_ZS_TASK2_V2	0.16	0.78	0.13	0.77	0.16
SINAL.task_2.PRM_ZS_TASK2_V3	0.10	0.86	0.05	0.83	0.11
Tomislav/Rowan_Task2.2.LLaMA2.13B.CHAT_1	0.01	0.61	0.00	0.00	0.00
Tomislav/Rowan_Task2.2.LLaMA2.13B.CHAT	0.01	0.33	0.00	0.00	0.00
UAMS_Task2-1_RareIDF	0.09	0.97	0.03	0.09	0.00
UboNLP_Task2.1_phi3-oneshot	0.54	0.65	0.32	0.37	0.00
unipd_t21t22_chatgpt	0.13	0.64	0.08	0.62	0.19
unipd_t21t22_chatgpt_mod1	0.22	0.52	0.20	0.60	0.18
unipd_t21t22_chatgpt_mod2	0.31	0.46	0.34	0.69	0.01

We also show the trend of the main results in Fig. 1. In the top row, the recall and precision graphs for all the terms and only the difficult terms are shown. In the bottom row, the BLEU score of the proposed definitions in relation with the recall and precision on difficult terms.

3.5 Analysis

The results shown in the previous subsection reveal that the use of large language models for the extraction of terms, the assessment of the difficulty of these terms, and the generation of the definitions to explain the difficult concepts are at an initial stage that will open new perspective in the Automatic Term Extraction panorama. In particular, compared to the recent results and surveys [see 7], the values of recall and precision are sufficiently good but suboptimal when compared to the state-of-the-art models (of course, we need to take into consideration that this is the first time participants dealt with this new dataset).

This concludes the results for the CLEF 2024 SimpleText Task 2: Complexity Spotting on Identifying and Explaining Difficult Concepts. Our main findings are the following: First, the runs submitted by the participants to this task are quite stable in terms

⁹ <https://cran.r-project.org/web/packages/sacRebleu/vignettes/sacReBLEU.html>.

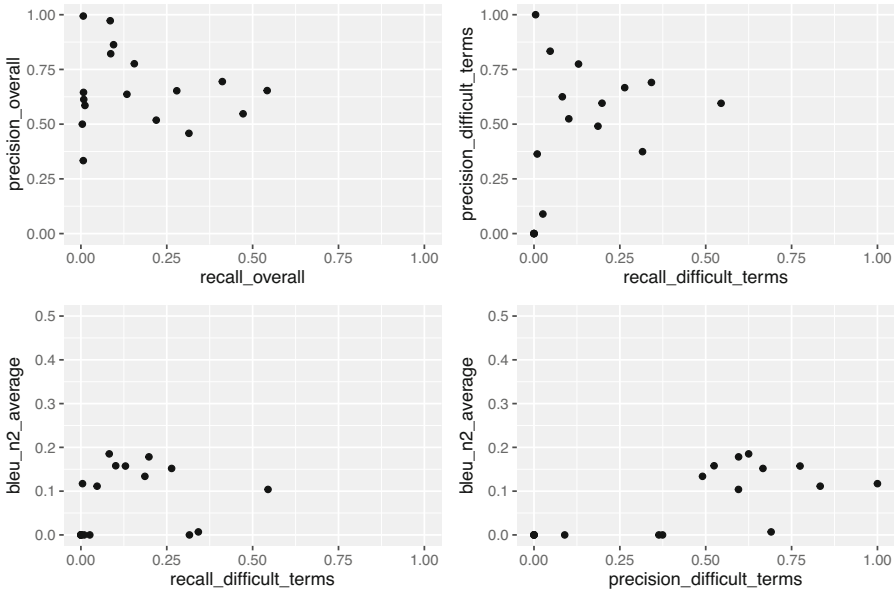


Fig. 1. Summary of Task 2 results.

of recall-precision performances when dealing with all the terms or the difficult ones. Independently from the difficulty of terms, the models proposed by the participants can achieve precision higher than .50 across a range of recall values. Second, the BLEU score of the generated definitions is also relatively stable ranging from 0.1 to 0.2 for any recall and precision values. Third, the best-performing runs are usually those that have some analysis of the optimal prompting or a manual interaction with the model. This is in line with the latest research studies on this issue [26]. We refer the reader to the CLEF 2024 SimpleText Task 2 Overview paper [31] for further details and discussion.

4 Task 3: Simplify Scientific Text

This section details *Task 3: Text Simplification* on Simplify Scientific Text.

4.1 Description

The goal of this task is to provide a simplified version of the sentences extracted from scientific abstracts. Participants will be provided with popular science articles and queries and matching abstracts of scientific papers, either split into individual sentences or as the entire abstracts.

Data. Task 3 uses the same corpus based on the sentences in high-ranked abstracts to the requests of Task 1. Our training data is a truly parallel corpus of directly simplified sentences (893 sentences this year) coming from scientific abstracts from the DBLP

Citation Network Dataset for *Computer Science* and Google Scholar and PubMed articles on *Health and Medicine*.

Other existing text simplification corpora used post-hoc aligned sentences [42, e.g.,]. The SimpleText corpus contains 893 directly simplified sentences, and a useful addition to existing high-quality corpora like NEWSELA [42] (2,259 sentences). Our track is the first to focus on the simplification of scientific text with a much higher text complexity than news articles.

Table 5. Example of SimpleText Task 3 reference versus input: deletions and insertions

Topic	Document	Output
G01.1	130055196	As various kinds <u>The rise of</u> output devices emerged , such as highresolution <u>like high-resolution</u> printers or a display of and PDA (Personal Digital Assistant) , <u>displays has increased the importance of need for high-quality resolution conversion has been increasing</u> . <u>This</u> The paper proposes a new method for enlarging image with to <u>make images bigger while maintaining high quality</u> . One of the largest problems on image enlargement <u>The main issue with enlarging images is the exaggeration of the jaggy that jagged edges can become exaggerated</u> . To remedy solve this problem , we propose suggest a new interpolation method , which uses artificial that helps us to estimate the value of the newly generated pixels using a neural network to determine the optimal values of interpolated pixels . The experimental experiment 's results are shown presented and evaluated analyzed . The <u>We evaluate the effectiveness of our methods is discussed by comparing with the conventional methods them to traditional approaches</u> .

Table 5 shows an example of a human reference simplification, combining the input sentences belonging to the abstract of the document $id = 130055196$ retrieved for query G01.1. Here, we show the deletions and insertions relative to the source input sentences (in this case in the first 4 sentences).

Available training data from 2023 includes 983 sentences from scientific abstracts plus manual simplifications [13]. These text passages were simplified either by master students in Technical Writing and Translation or by a domain expert (a computer scientist) and a professional translator (native English speaker) working together. The new sentence-level evaluation (test) data in 2024 consists of 578 sentences.

In 2024, we expanded the training and evaluation data. In addition to sentence-level text simplification, we will provide passage-level input and reference simplifications, with the train data corresponding to 175 abstracts with corresponding human simplifications. The new abstract-level evaluation (test) data in 2024 consists of 103 abstracts.

Evaluation. In 2024, we emphasize large-scale automatic evaluation measures (SARI, BLEU, compression, readability) that provide a reusable test collection. This automatic evaluation will be supplemented with a detailed human evaluation of other aspects, essential for deeper analysis. Almost all participants used generative models for text

simplification, yet existing evaluation measures are blind to potential hallucinations with extra or distorted content [13]. In 2024, we provide further analysis of ways to detect and quantify spurious content in the output, potentially corresponding to what is informally called “hallucinations”.

4.2 Participant’s Approaches

A total of 15 teams submitted 83 runs in total.

AB/DPV. Varadi and Bartulović [40] submitted one run for Task 3. Their approach is an LSTM model for the sentence-level task.

Sharigans. Ali et al. [2] submitted a total of two runs for Task 3. Their approach is a GPT-3.5 model for both the sentence-level and abstract-level tasks.

Tomislav/Rowan. Mann and Mikulandric [27] submitted a total of two runs for Task 3. Their approach is the LLama 2 model with a range of prompts and post-processing for both the sentence-level and abstract-level tasks.

Petra/Diana. Elagina and Vučić [10] submitted one run for Task 3. Their approach is a LLaMA model for the sentence-level task.

Dajana/Katya submitted one run for Task 3. Their approach which follows standard text simplification approaches is applied to the sentence-level task.

AIIRLab Largey et al. [25] submitted a total of eight runs for Task 3. Their approach uses LLaMA3 and Mistral models with different prompting and fine-tuning, for both the sentence-level and abstract-level tasks.

UBO Vendeville et al. [41] submitted a total of four runs for Task 3. Their approach is to prompt a smaller Phi3 model for lexical and grammatical text simplifications, for both the sentence-level and abstract-level tasks.

UAmsterdam Bakker et al. [4] submitted a total of ten runs for Task 3. They experiment with GPT-2, and Wiki and Cochrane-trained models at the sentence, paragraph, and document-level text simplification, for both sentence-level and document-level tasks.

UZHPandas Michail et al. [28] submitted a total of ten runs for Task 3. They experiment with a multi-prompt Minimum Bayes Risk (MBR) decoding approach to the sentence-level task.

Elsevier Capari et al. [5] submitted a total of ten runs for Task 3. Their approach is based on a GPT-3.5 model experimenting with zero-shot and few-shot prompts for both sentence-level and abstract-level tasks.

Frane/Andrea submitted one run for Task 3. Their approach which follows standard text simplification approaches is applied to the sentence-level task.

Arampatzis. (No paper received) submitted a total of eight runs for Task 3. Their approach is a range of models (DistilBERT, T5) for both the sentence-level and abstract-level tasks.

Ruby. (No paper received) submitted two runs for Task 3. Their approach uses standard models for both sentence-level and abstract-level tasks.

SONAR. (No paper received) submitted a single run for Task 3. Their approach is a standard model for the sentence-level task.

PiTheory. (No paper with run details received) submitted a total of twenty runs for Task 3. Their approach uses pre-trained BART and T5 models but contains very few results for both the sentence-level and abstract-level tasks.

4.3 Results

This section details the results of the task, for both sentence-level and abstract-level test simplification subtasks.

Table 6. Results for CLEF 2024 SimpleText Task 3.1 sentence-level text simplification (task number removed from the run_id) on the test set

run_id	count	FKGL	SARI	BLEU	Compression ratio	Sentence splits	Levenshtein similarity	Exact copies	Additions proportion	Deletions proportion	Lexical complexity score
Source	578	13.65	12.02	19.76	1	1	1	1	0	0	8.8
Reference	578	8.86	100	100	0.7	1.06	0.6	0.01	0.27	0.54	8.51
Elsevier_run1	578	10.33	43.63	10.68	0.87	1.06	0.59	0.00	0.45	0.53	8.39
Elsevier_run4	577	11.73	43.14	12.08	0.85	1.00	0.63	0.00	0.37	0.50	8.54
Elsevier_run8	577	12.40	42.95	12.35	0.90	1.02	0.63	0.00	0.35	0.50	8.66
Elsevier_run6	577	12.65	42.88	11.76	0.95	1.00	0.64	0.00	0.38	0.47	8.63
Elsevier_run7	577	12.55	42.87	12.20	0.87	1.00	0.63	0.00	0.35	0.51	8.67
Elsevier_run9	577	12.53	42.61	12.15	0.87	1.00	0.63	0.00	0.35	0.50	8.67
Elsevier_run3	577	11.50	42.58	15.75	0.76	0.98	0.68	0.00	0.23	0.46	8.68
Elsevier_run10	577	12.57	42.49	11.91	0.91	1.02	0.63	0.00	0.34	0.50	8.67
AIIRLab.Jlama-3-8b_run1	578	8.39	40.58	7.53	0.90	1.37	0.56	0.00	0.48	0.58	8.45
AIIRLab.Jlama-3-8b_run3	578	9.47	40.36	6.26	1.17	1.52	0.53	0.00	0.53	0.56	8.51
AIIRLab.Jlama-3-8b_run2	578	10.33	39.76	5.46	1.03	1.19	0.51	0.00	0.60	0.56	8.34
UZH.Pandas_simple_cot	578	13.74	39.59	3.38	3.44	2.67	0.41	0.00	0.76	0.12	8.61
UZH.Pandas_simple	578	11.24	39.28	5.67	0.88	0.98	0.52	0.00	0.53	0.62	8.45
Sharengans_finetuned	578	11.39	38.61	18.18	0.83	1.07	0.77	0.11	0.16	0.32	8.70
UZH.Pandas_selection_sle_cot	578	6.49	38.38	1.03	4.76	6.26	0.30	0.00	0.89	0.14	8.30
UZH.Pandas_selection_inter.def	578	21.36	38.29	3.13	1.93	0.99	0.46	0.00	0.69	0.33	8.86
UZH.Pandas_selection_lens.cot	578	6.74	38.16	1.10	4.54	5.88	0.32	0.00	0.87	0.14	8.32
UZH.Pandas_SY_target_cot	578	6.39	37.95	0.97	4.73	6.25	0.30	0.00	0.89	0.14	8.30
UZH.Pandas_selection_lens	578	21.29	37.79	2.71	1.97	1.01	0.44	0.00	0.71	0.34	8.85
UBO_PhiMini-s	578	8.74	36.78	0.58	18.23	23.48	0.47	0.00	0.66	0.29	8.89
UZH.Pandas_selection_lens_l	578	7.79	36.72	3.65	0.72	0.98	0.46	0.00	0.54	0.73	8.25
UBO_PhiMini-sl	578	6.16	36.53	0.61	6.92	9.81	0.38	0.00	0.80	0.42	8.72
UZH.Pandas_SY_target_inter.def	578	19.30	36.53	2.27	1.76	1.01	0.45	0.00	0.70	0.41	8.87
UZH.Pandas_selection_sle	578	6.07	35.30	2.57	0.65	0.98	0.43	0.00	0.56	0.78	8.17
UZH.Pandas_SY_target	578	5.94	34.91	2.29	0.66	0.99	0.43	0.00	0.57	0.78	8.17
RubyAIYoungTeam	578	8.76	34.40	15.37	0.60	1.22	0.69	0.03	0.05	0.44	8.71
SONAR_SONARonlinereg	578	13.14	32.12	18.41	0.97	1.01	0.93	0.13	0.11	0.13	8.73
UAMS_GPT2_Check	578	11.47	29.91	15.10	1.02	1.23	0.87	0.14	0.17	0.14	8.68
UAMS_GPT2	578	10.91	29.73	13.07	1.30	1.50	0.79	0.06	0.29	0.12	8.63
Arampatzis_T5	578	13.18	28.92	10.66	1.12	1.10	0.72	0.03	0.34	0.37	9.06
UAMS-Wiki.BART_Snt	578	12.13	27.45	21.56	0.85	0.99	0.89	0.32	0.02	0.16	8.73
Arampatzis_DistilBERT	578	5.85	19.00	13.56	1.03	3.00	0.95	0.00	0.22	0.11	8.65
UAMS_Cochrane.BART_Snt	578	13.22	18.45	19.21	0.95	0.99	0.96	0.59	0.02	0.07	8.77
Arampatzis_METHOD	578	13.65	12.12	19.77	1.00	1.00	1.00	0.99	0.00	0.00	8.80

4.4 Task 3.1: Sentence-Level Scientific Text Simplification

Table 6 shows the Task 3.1 (sentence-level text simplification) results. The table is restricted to submissions covering a sufficient number of input sentences. We show a number of evaluation scores against the human reference simplifications, in particular SARI and BLEU. In addition, we provide additional text statistics on the system output such as FKGL, and a comparison to the source input.

We make a number of observations. First, the table is sorted on SARI, the main automatic text simplification measure used in the track. We observe SARI scores of 30+ % for the majority of systems and 40+ % for the top-scoring systems. This high overlap with the human reference simplifications is encouraging and indicates that the effectiveness of text simplification approaches, traditionally trained on youth news reading corpora like Newsela, also extends to scientific text.

Second, in terms of the level of text complexity, readability measures like FKGL provide a rough indicator of lexical and grammatical complexity. The original sentences have an FKGL of 13–14 corresponding to university-level text, and the majority of systems reduce this to an FKGL of 11–12 corresponding to the exit level of compulsory education. This is an encouraging result, as it indicates that the scientific text simplification approach can be a viable approach to lower the textual complexity of scientific text toward the range acceptable by a layperson. Although this is positive indicator, this approximate measure does not take into account terminological complexities as studied in Task 2, or ways to retrieve all and only more accessible abstracts in Task 1 [14].

Third, the table includes various other scores that indicate that there is still considerable room for improvement in scientific text simplification. Throughout the table the BLEU evaluation measure remains very low, and leads to a different ranking of systems with some of the best systems on BLEU demonstrating superior overlap with the human reference simplifications. The table also reveals some runs with very high “compression” ratios and sentence splits, as well as high proportions of additions. While evaluation measures like SARI are essential for understanding important aspects of text simplification output quality, they are also known to be relative insensitive to content outside the intersection with the manual text simplifications. Hence high levels of insertion of content can still lead to favorable SARI scores, and even improve text statistics like FKGL, without conveying key content of the original text.

4.5 Task 3.2: Abstract-Level Scientific Text Simplification

Table 7 shows the Task 3.2 (abstract-level text simplification) results. Again we restrict the table to submissions covering a sufficient number of input abstracts.

Table 7. Results for CLEF 2024 SimpleText Task 3.2 abstract-level text simplification (task number removed from the run_id) on the test set

run_id	count	FKGL	SARI	BLEU	Compression ratio	Sentence splits	Levenshtein similarity	Exact copies	Additions proportion	Deletions proportion	Lexical complexity score
Source	103	13.64	12.81	21.36	1	1	1	0	0	0	8.88
Reference	103	8.91	100	100	0.67	1.04	0.6	0	0.23	0.53	8.66
AIIRLab_Jlama-3-8b_run1	103	9.07	43.44	11.73	1.01	1.38	0.51	0.00	0.37	0.56	8.57
AIIRLab_Jlama-3-8b_run3	103	10.17	43.21	11.03	1.15	1.47	0.52	0.00	0.40	0.51	8.66
Elsevier_run2	103	11.01	42.47	10.54	1.04	1.22	0.51	0.00	0.38	0.55	8.60
AIIRLab_Jlama-3-8b_run2	103	10.22	42.19	7.99	1.31	1.38	0.48	0.00	0.53	0.52	8.44
Elsevier_run5	103	12.08	42.15	10.96	1.04	1.15	0.52	0.00	0.36	0.53	8.75
Sharingans_finetuned	103	11.53	40.96	18.29	1.20	1.39	0.65	0.00	0.24	0.34	8.80
UBO_Phi4mini-ls	103	8.45	38.79	5.53	1.21	1.75	0.43	0.00	0.40	0.63	8.53
UBO_Phi4mini-l	103	9.96	38.41	10.01	1.29	2.11	0.55	0.00	0.24	0.51	9.03
UAms_GPT2_Check_Abs	103	12.85	36.47	13.12	0.91	0.92	0.59	0.00	0.18	0.45	8.73
UAms_Cochrane_BART_Doc	103	14.46	33.51	9.39	0.65	0.58	0.54	0.04	0.06	0.53	8.80
UAms_Cochrane_BART_Par	103	16.53	31.58	15.40	1.08	0.80	0.67	0.04	0.15	0.32	8.81
UAms_GPT2_Check_Snt	103	11.57	30.71	15.24	1.54	1.70	0.78	0.00	0.27	0.13	8.77
Arampatzis_METHOD	103	0.00	28.28	0.00	0.00	0.00	0.00	0.00	0.00	1.00	10.82
Arampatzis_T5	103	0.00	28.28	0.00	0.00	0.00	0.00	0.00	0.00	1.00	10.82
Arampatzis_DistilBERT	103	0.00	28.28	0.00	0.00	0.00	0.00	0.00	0.00	1.00	10.82
UAms_Wiki_BART_Doc	103	15.68	26.50	15.11	1.51	1.14	0.76	0.01	0.25	0.11	8.79
UAms_Wiki_BART_Par	103	13.11	23.92	19.49	1.39	1.37	0.81	0.01	0.11	0.10	8.86

We make a number of observations. First, in terms of evaluation measures like SARI we see again similar encouraging performance levels when evaluating against the human reference simplifications. This is partly due to the use of proven sentence-level text simplification models with the output merged back into the entire abstract. Second, there remains room for improvement in capturing the human simplifications more closely, as the BLEU score remains low throughout. Here, the more conservative approaches seem to obtain better scores. Third, we see less extreme values on the other indicators, but still considerable variation in the compression ratio and number of splits, and proportions of addition and deletions. We will investigate how much of the output is grounded in the source sentences and abstracts below.

Many submissions rely on proven sentence-level text simplification approaches, with results closely mirroring those observed for the sentence-level task. It is encouraging to see solid performance for the approaches that perform text simplification at the entire abstracts in one pass. This holds the promise to incorporate the discourse structure, use more complex text simplifications operations such as deletions and merges, and deploy planner-based approaches to the text simplification of long documents.

4.6 Analysis

We conduct a deeper analysis of how much of the generated simplified output sentences and abstracts can be traced to the source input. In particular, we look at spurious generated content and it's prevalence in the submitted generated text simplifications. This content is at risk of being introduced gratuitously by the generative model, and what is informally referred to as "hallucinations."

Table 8. Example of SimpleText Task 3 output versus input: deletions, insertions, and whole sentence insertions

Topic	Document	Output
G01.1	130055196	As various kinds of output devices emerged , such as highresolution printers or a display of PDA (Personal Digital Assistant) , the . <u>The importance of high-quality resolution conversion has been increasing .</u> This paper proposes a new method for enlarging <u>an</u> image with high quality . <u>It will involve using a combination of high-speed imaging and high-resolution video.</u> One of the largest <u>biggest</u> problems on image enlargement is the exaggeration of the jaggy edges . <u>This is especially true when the image is enlarged , as in this case.</u> To remedy this problem , we propose a new interpolation method , which . <u>This method</u> uses artificial neural network to determine the optimal values of interpolated pixels . The experimental results are shown and evaluated . <u>The results are compared to other studies and found to be inconclusive.</u> The effectiveness of our methods is discussed by comparing with the conventional methods . <u>Our methods are designed to help people with mental health problems , not just as a way to cure them.</u>

Earlier in Table 5, we showed an example of a human reference simplification, combining the input sentences belonging to the abstract of the document $id = 130055196$ retrieved for query G01.1. We can do the same for the automatically generated scientific text simplifications. We show again the deletions and insertions relative to the source input sentences. Table 8 shows an example output simplification of one of the participating teams, for the same input sentences as in Table 5 above. Most simplifications are revisions of the input, but we also observe that sometimes an entire sentence is inserted (shown as xxx in Table 8).

We provide a detailed analysis quantifying the prevalence of spurious content in the CLEF 2024 SimpleText Task 3 Overview paper [17]. We re-aligned the generated output with the original source sentences, and flag here only entire output sentences that do not share a single token with the input. The example in Table 8 is an extreme case picked to illustrate both the importance and complexity of detecting such spurious content. However, our analysis reveals that the amount of spurious content is varying but far from infrequent. A total of 17 out 36 submissions (47%) have spurious whole sentences in at least 10% of the input sentences. In fact, 14 (39%) submissions in at least 20% of the input, and 7 (19%) submissions in at least 50% of the input sentences. The detection of non-aligned output sentences is indicative but imperfect. For example, significant reordering of content may lead to false positives in rare cases, and unusual tokenization or formatting may affect the alignment with the source even systematically. Note also that the detected additions may introduce helpful background knowledge or other useful information to contextualize the information in the source sentences.

We make a number of observations based on our analysis in this section. First, the fraction of sentences with spurious content is very low for some submissions, however, for other submissions, the fraction is very substantial. Second, the standard evaluation measures used for text simplification, and in fact for any text generation task in NLP, do not take this aspect into account. A submission with significant spurious content can still obtain very high text overlap with the reference, and hence obtain a very high performance score. Third, and more generally, human evaluation and this type of analysis feel crucial to accurately evaluate generative models for the NLP and IR challenges addressed in our Track and in CLEF in general.

This concludes the results for the CLEF 2024 SimpleText Task 3: Text Simplification on Simplify Scientific Text. Our main findings are the following: First, we observe competitive performance for scientific text simplification, both on evaluation against the human reference simplifications and on text statistics such as FKGL readability score. Second, the abstract-level text simplification results is a mixture of sentence-level and passage-level text simplification approaches. Third, our analysis reveals a very high and varying range of spurious text generation, not detected by standard evaluation measures, and a major concern in the use of these model in a real-world setting. More generally, almost all participants use generative models (for the task, the track, and CLEF in general), and the track offers a unique setting to study some of the inherent limitations of generative models. We refer the reader to the CLEF 2024 SimpleText Task 3 Overview paper [17] for further details and discussion.

5 Task 4: Tracking the State-of-the-Art in Scholarly Publications

This section details *Task 4: SOTA?* on Tracking the State-of-the-Art in Scholarly Publications.

5.1 Description

In Artificial Intelligence (AI), a common research objective is the development of new models that can report state-of-the-art (SOTA) performance. The reporting usually comprises four integral elements: Task, Dataset, Metric, and Score. These (Task, Dataset, Metric, Score) tuples coming from various AI research papers go on to power leaderboards in the community. Leaderboards, akin to scoreboards, traditionally curated by the community, are platforms displaying various AI model scores for specific tasks, datasets, and metrics. Examples of such platforms include the [benchmarks feature](#) on the [Open Research Knowledge Graph](#) and [Papers with Code](#) (PwC). Utilizing text mining techniques allows for a transition from the conventional community-based leaderboard curation to an automated text mining approach. Consequently, the goal of Task 4: SOTA? is to develop systems which given the full text of an AI paper, are capable of recognizing whether an incoming AI paper indeed reports model scores on benchmark datasets, and if so, to extract all pertinent (Task, Dataset, Metric, Score) tuples presented within the paper.

Data. The training and test datasets for this task are derived from community-curated (T, D, M, S) annotations for thousands of AI articles available on PwC (CC BY-SA). We will utilize the dataset obtained from our prior work, specifically the PwC source downloaded on May 10, 2021 [20,23], which comprised over 7,500 articles. These articles, originally sourced from arXiv under CC-BY licenses, are available in TEI XML format, each accompanied by one or more (T, D, M, S) annotations from PwC. While our previous work employed dataset splits for two-fold cross-validation experiments, for the SimpleText Task 4, we will establish new 70/30 train/test splits, providing approximately 5,000 annotated articles for participant training. A preliminary version of our training dataset can be accessed on Github <https://github.com/jd-coderepos/sota>.

The test set will strategically include only those articles with TDMs seen in the training set, creating a few-shot evaluation setting. Furthermore, in our subsequent research [21], we explored a zero-shot evaluation setting, wherein the dataset contained articles with at least one T, D, or M not seen in the model’s training set. Thus in addition to the few-shot evaluation, we intend to introduce a second evaluation setting for Task 4, evaluating models in a zero-shot context, for which a new test dataset will be created. Finally, ongoing efforts involve expanding the primary task corpus by incorporating approximately 1,500 articles into both the train and test sets that do not report leaderboards. These articles will be annotated with the *unknown* label. Consequently, systems developed in our shared task will have comprehensive applicability to any AI article, extracting (T, D, M, S) annotations for articles that contain them and assigning *unknown* for those that do not.

Evaluation. As discussed above, in Task 4 participant systems will be evaluated in the two evaluation settings. For **Few-shot** evaluation, trained systems will have to predict

(T, D, M, S) annotations on a new collection of articles' full-text. The labels in the gold dataset will include only (T, D, M, S)'s seen at least once in training. For **Zero-shot** evaluation, the task is as above with a different collection of articles, which have (T, D, M, S) with unseen T, D, or M in the training set. In both settings, the standard recall, precision, and F-score metrics will be used to report scores to the participant systems.

5.2 Participant's Approaches

A total of 2 teams submitted 36 runs in total.

AMATU by Staudinger et al. [36] submitted a total of three runs for the **few-shot** evaluation phase of Task 4. They submitted nine runs for the **zero-shot** evaluation phase of Task 4. Their general approach to extract the (T, D, M, S) annotations were in two main categories: 1) a pure pattern-based approach inspired after AxCell [24], and 2) an AI-based approach using LLMs with a zero-shot prompt and a few-shot prompt tested for GPT-3.5 [32] and Mistral-7B [19]. For the latter category, they also experimented with variants on the input scholarly article text from which the (T, D, M, S) annotations were expected to be extracted. This we generally refer to as the *context*. They tried two context variants: 1) full paper text and 2) only the text from sections referring to experiments and results, in addition to the abstract, which was pre-extracted inspired by the Argumentative Zoning (AZ) method [38].

L3S by Kabongo et al. [22] submitted a total of 12 runs for the **few-shot** evaluation phase of Task 4. They submitted 12 runs for the **zero-shot** evaluation phase of Task 4. Their approach entailed leveraging the FLAN-T5 [6] strategy which encompassed fine-tuning a pre-trained LLM with a standard set of instructions to better equip them to handle various tasks. Leveraging the applicable instructions from the FLAN-T5 collection, they fine-tuned LLMs, viz. Mistral-7B and LLaMA 2 [39], to make them better suited to handle the (T, D, M, S) extraction task. Furthermore, they also tested the most recent proprietary GPT models viz. GPT-4 [1] and GPT-4o. Finally, as the information extraction context they tried 3 different methods: DocTAET ((T)-title, (A)- abstract, (E)-experimental setup, and (T)-tabular information parts of the full-text), DocREC (text selected from the sections named (R)-results, (E)-experiments, and (C)-conclusions), and DocFULL (full paper text). Resultingly, for each evaluation phase they submitted a total of 4 models x 3 contexts = 12 runs.

5.3 Results

Table 9 and Table 10 present a summary of the results from the two teams. Overall, given *Team AMATU*'s approaches, the pattern-based method proved a competitive solution to the SOTA challenge, in comparison to advanced LLM-based solutions. While the LLM solution did outperform the pattern-based approach the difference was minor. Furthermore, comparing the zero-shot versus few-shot paradigms, the LLMs were significantly more effective in the few-shot setting i.e. when shown successful task completion outputs. Also, the LLM performed significantly better when given the full paper text as input from which to extract (T, D, M, S) as opposed to given selective text using the AZ method.

Table 9. Evaluation results for the binary classification or filtering of papers with and without leaderboards (reported as General Accuracy) and as a structured summary generation task (reported with ROUGE metrics). *Team AMATU*’s few-shot evaluation results are reported for AxCell and their zero-shot evaluation results are reported for GPT-3.5 via the few-shot prompting paradigm. *Team L3S*’s results are reported for Mistral-7B finetuned with the DocTAET context. The best results are shown in bold.

	Few-shot					Zero-shot				
	Rouge				Gen.	Rouge				Gen.
	1	2	L	Lsum	Acc.	1	2	L	Lsum	Acc.
<i>AMATU</i>	58.34	12.98	57.34	54.4	75.59	73.72	6.07	72.72	72.57	85.93
<i>L3S</i>	57.24	19.67	56.28	56.19	89.68	73.54	12.23	73.01	72.95	95.97

Table 10. Evaluation results w.r.t. the individual (Task, Dataset, Metric, Score) elements and Overall in terms of **F1 score**. *Team AMATU*’s few-shot evaluation results are reported for AxCell and their zero-shot evaluation results are reported for GPT-3.5 via the few-shot prompting paradigm. *Team L3S*’s results are reported for Mistral-7B finetuned with the DocTAET context. The best results are shown in bold.

Model	Mode	Few-shot					Zero-shot				
		T	D	M	S	Overall	T	D	M	S	Overall
<i>AMATU</i>	Exact	27.11	23.22	24.85	9.34	21.13	10.01	13.16	11.65	9.85	11.16
	Partial	28.08	24.92	25.8	10.86	22.62	16.12	17.12	13.72	11.1	14.52
<i>L3S</i>	Exact	33.38	18.51	24.23	1.87	19.50	26.99	14.32	22.04	1.20	16.14
	Partial	46.35	32.75	34.16	2.25	28.88	44.90	27.29	32.23	1.41	26.46

For *Team L3S*, in both the evaluation phases, their model results showed that minimal finetuning of relatively smaller LLMs, specifically Mistral-7B, equips them for (T, D, M, S) extraction task surpassing the performance of LLMs, specifically the latest GPT-4 proprietary models, with a significantly more vast parameter space. The overall best results even for the extraction of the (T, D, M, S) elements was obtained by Mistral given the DocTAET context.

Comparing *Team AMATU* and *Team L3S*, none of the systems from the former team were finetuned to the task. Thus *Team AMATU* presents novel insights into the community to leveraging LLM’s effectively for the (T, D, M, S) extraction objective using clever prompt engineering strategies that shows comparable performans to the latter teams’ computationally intensive finetuning approach. It maybe that finetuning would be essential to create the most optimal model, however, from the team’s solutions the importance prompt engineering for effective downstream performance is clearly emphasized.

This concludes the results for the CLEF 2024 SimpleText Task 4: SOTA? on Tracking the State-of-the-Art in Scholarly Publications. Our main findings are the following: First, effective prompting paradigms should be a go-to strategy to test LLMs out-of-the-box for the SOTA? shared task objective. Second, finetuning small-scale models makes them better able to handle the SOTA? objective than larger-scale LLMs known for their

generative AI abilities when simply applied to the IE task. Third, the paper context over which the IE task is expected to be performed must have an ideal balance of length versus selectivity of specific sections in the paper that indeed are highly likely to contain mentions of the (T, D, M, S). On the extreme end of the spectrum, using the full paper text without effective context selection hinders and seems to distract the LLM downstream IE task performance. We refer the reader to the CLEF 2024 SimpleText Task 4 Overview paper [9] for further details and discussion.

6 Conclusions

This paper described the setup of the CLEF 2024 SimpleText track, which contains four interconnected tasks on scientific text summarization and simplification. More detailed discussion on each of the tracks can be found in the task overview papers: Task 1 on *Content Selection* [35]; Task 2 on *Complexity Spotting* [31]; Task 3 on *Text Simplification* [17]; and Task 4 on *SOTA?* [9].

The main aim of our track, and the CLEF evaluation forum as a whole, is i) to construct corpora and evaluation resources to stimulate research on scientific text summarization and simplification, and ii) to foster a community of IR, NLP, and AI researchers working together on the important task of making science more accessible for everyone.

Within the CLEF 2024 SimpleText track, we have constructed extensive corpora and manually labeled evaluation data. First, a large corpus of over 4 million scientific abstracts that can be used for popular science search requests, with corresponding relevance judgments. Second, scientific terms from sentences coming from scientific abstracts with manually attributed difficulty scores and terminology expert provided explanations and definitions. Third, a parallel corpus of manually simplified sentences and abstracts from the scientific literature. Fourth, a corpus for extracting key performance indicators on standard benchmarking data from full text scientific papers. These reusable corpora and evaluation resources are available to participants and other researchers who want to work on the important problem of making scientific information open and easily accessible for everyone.

In terms of a building a community researching scientific text summarization and simplification, the track saw a record attendance in 2024: with a fourth information extraction task added, more runs submitted with the largest number of participating teams ever. In fact, the community is broadening beyond CLEF and raising general interest in generative scientific text summarization and simplification [30].

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